PRODUCT ANALYTICS MANAGER — ADVANCED CASE STUDY

Personalization, Recommendation Quality & Retail Media Monetization

Target roles: Product Manager / Analytics Manager — Amazon, Flipkart, Meta-style marketplace environments

Business Context

The marketplace has invested heavily in recommendation systems and a fast-growing retail-media business. " "Leadership believes personalization is increasing conversion, but category leaders are concerned that recommendation " "models may over-optimize clicks while reducing discovery quality, customer trust and long-term value. At the same time, " "advertising revenue is growing rapidly, creating pressure to place more sponsored products into high-traffic surfaces.

Your Mandate

Act as the Product Analytics Manager reporting to the VP of Product. Determine whether the company should scale the " "recommendation platform and retail-media business, redesign the experience, or impose guardrails. You must connect " "recommendation quality, customer behavior, conversion, margin, advertising economics and customer experience rather than " "optimizing any single KPI.

Core Executive Question

“Clicks and ad revenue are increasing. How do you prove that personalization is creating durable customer and business value rather than simply extracting more clicks?”

Available Data

Table	Approx. rows	Purpose
Customers	5,000+	Customer profile, segment, city and acquisition
Products	1,400	Category, price, seller and rating
Recommendations_Impressions	32,000+	Recommendation exposure, rank, model version and clicks
Customer_Events	28,000+	Views, clicks, wishlist, cart and purchase behavior
Orders	23,500+	GMV, discount, margin, source and repeat behavior
Ad_Events	5,200+	Ad impressions, clicks, spend and attributed GMV
Campaigns	180	Advertiser, bidding strategy, target ROAS and budget
Recommendation_Feedback	7,200+	Relevance, ad-load and repeated-item feedback

Data Is Intentionally Dirty

Do not assume the files are analysis-ready. Identify and document duplicates, missing values, inconsistent category/surface " "labels, mixed casing and whitespace, invalid ranks, impossible click values, negative dwell times, negative GMV/spend, " "unrealistic margins, duplicated events and other quality problems. Your data-quality decisions must be defensible because they " "can materially change business conclusions.

Required Analysis — Excel

• Create a data-quality audit and a clean analytical layer. Quantify the impact of each major cleaning rule.

• Build recommendation funnel metrics: impressions → clicks → downstream product actions → orders.

• Compare recommendation performance by model version, surface, rank position, category, customer segment and city.

• Calculate CTR, conversion rate, revenue/order, contribution margin and repeat-order indicators.

• Evaluate whether higher recommendation CTR actually translates into higher downstream value.

• Measure recommendation concentration: percentage of impressions/orders/revenue generated by the top products.

• Build customer-level segments using engagement, recommendation response and purchase behavior.

• Evaluate advertising economics: CTR, CPC, ROAS, spend, attributed GMV and margin contribution.

• Create a scenario model for allocating an additional ■30 crore across recommendation infrastructure, experimentation, ad inventory, relevance improvements and customer-experience safeguards.

• Show sensitivity of the recommendation and advertising investment case under conservative, base and aggressive assumptions.

Required Analysis — SQL

• Use at least 5 tables in meaningful joins.

• Use CTEs to construct a customer-level recommendation-to-purchase funnel.

• Use window functions to rank products within category, customers within segments and campaigns within advertiser groups.

• Identify the relationship between recommendation rank position and downstream conversion.

• Compare model versions using normalized KPIs rather than raw volume.

• Find customers repeatedly exposed to the same products and quantify whether repetition is associated with lower engagement.

• Calculate customer-level recommendation-assisted GMV and contribution margin.

• Identify campaigns with high ROAS but weak margin contribution and campaigns with strong incremental-looking economics.

• Create risk/quality bands for recommendation experiences based on CTR, conversion, margin and negative feedback.

• Produce an executive table showing where incremental investment should be made and where it should be restricted.

Required Analysis — Power BI

• Executive Scorecard — GMV, recommendation contribution, conversion, margin, ad revenue, ROAS and customer-experience indicators.

• Recommendation Performance — model version, surface, rank, category and customer segment.

• Customer & Discovery — engagement segments, repeat behavior, discovery concentration and recommendation-assisted purchase behavior.

• Retail Media — spend, clicks, CTR, CPC, ROAS, attributed GMV and margin by advertiser/category/bid strategy.

• Investment Simulator — ■30 crore allocation, expected impact, scenario assumptions and KPI trade-offs.

Data Strategy Challenge

Design a practical analytics/data strategy for the next 12 months. Your strategy should explain the target data architecture, " "event taxonomy, customer/product identity resolution, recommendation measurement framework, experimentation approach, data-quality " "controls, feature/metric governance, model monitoring, incrementality measurement and executive KPI ownership.

• Define the minimum event instrumentation required to measure recommendation quality end-to-end.

• Propose a single source of truth for customer, product, order, recommendation and advertising metrics.

• Design a recommendation KPI tree separating leading indicators from business outcomes.

• Define guardrail metrics so recommendation optimization does not damage customer experience or margin.

• Propose an A/B testing and incrementality framework for both recommendations and sponsored placements.

• Define model/data drift monitoring and alert thresholds.

• Explain how Product, Data Science, Engineering, Category and Ads teams should share metric ownership.

Executive Recommendation

Prepare a 5–7 slide executive recommendation answering: Should the company invest the next ■30 crore in personalization and " "retail media? Your answer must include the evidence, trade-offs, recommended allocation, expected business impact, major risks, " "guardrails and the first 90 days of execution.

Interview / Viva Questions

• Your recommendation CTR increased 30%. Why might this be a bad result?

• How would you distinguish correlation from causal impact in recommendation-assisted orders?

• Why can a high-ROAS advertising campaign still destroy value?

• How would you decide whether to show a sponsored product or an organic recommendation?

• What would you optimize if the CEO asked for GMV growth but the CFO asked for contribution margin?

• How would you measure long-term discovery quality rather than short-term clicks?

• What data would you demand before approving the ■30 crore investment?

• If your model performs best on clicks but worst on repeat purchase, which KPI wins and why?

Final Deliverables

• Clean analytical dataset + data-quality log

• Excel business analysis and ■30 crore investment model

• SQL analysis script covering multi-table joins, CTEs and window functions

• Power BI executive dashboard

• 5–7 slide executive recommendation

• 2–3 page data strategy

• One-page portfolio/resume summary describing the business impact and analytical approach

Portfolio positioning: “Designed and evaluated a marketplace personalization and retail-media analytics strategy by linking recommendation quality, customer behavior, contribution margin and advertising economics; built an investment framework for ■30 crore.”